

Homework Submission

Student Name
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Problem Statement

$$x^2 + 3x + 2 = 0 \tag{1}$$

The task is to solve the quadratic equation $x^2 + 3x + 2 = 0$.

Solution Method: Factoring

To solve the equation, we look for two numbers that multiply to the constant term (2) and add up to the coefficient of the linear term (3). These numbers are 1 and 2.

Thus, the quadratic expression can be factored as:

$$(x + 1)(x + 2) = 0 \tag{2}$$

Setting each factor to zero gives the solutions:

- $x + 1 = 0 \implies x = -1$
- $x + 2 = 0 \implies x = -2$

Final Answer

The solutions to the equation are $x = -1$ and $x = -2$.